



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pathology of Soft Tissue

Principles and Practice of Osteopathic Medicine I

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Make.
Heal.
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Session Objectives

- 1. Overview of soft tissue **tumors** and review of normal with description and discussion of **lipoma, liposarcoma** and its **variants**
- 2. Describe and discuss the morphological and pathological features of fibrous proliferations and tumors such as **nodular fasciitis, superficial and deep fibromatoses, and fibrosarcoma**
- 3. Describe and discuss the morphological and pathological features of **rhabdomyosarcoma and its variants**
- 4. Describe and discuss the morphological and pathological features of **synovial sarcoma**
- 5. Describe and discuss the morphological and pathological features of **undifferentiated pleomorphic sarcoma**

Soft tissue tumors

- Overview/Introduction
- Brief review of normal histology soft tissue cells
- **Adipocytic**
 - Lipoma
 - Liposarcoma
- **Fibroblastic**
 - Nodular fasciitis
 - Fibromatosis – Superficial and Deep
 - Fibrosarcoma
- **Skeletal muscle**
 - Rhabdomyosarcoma
- **Unknown origin**
 - Synovial Sarcoma
 - Undifferentiated Pleomorphic Sarcoma

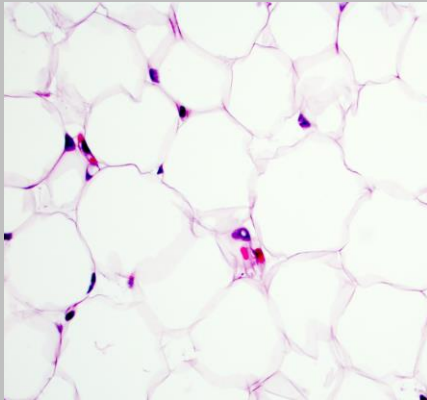
SOFT TISSUE TUMORS

- **Malignant** soft tissue tumors incidence is less than 1% of all cancers; causes 2% of cancer mortality – **usually aggressive**
- **Benign > malignant**; 100 to 1 (except for skeletal muscle neoplasms)
- Most occur in extremities, esp. thigh; 15% arise in children
- Commonly present as a growing mass, sometimes painful, possible loss of range of motion
- Most sporadic; minority associated with mutations in tumor suppressor genes associated with syndromes (ex., Neurofibromatosis 1, Gardner syndrome, Li-Fraumeni, etc)
- Some linked to environment – radiation, burns, toxins
- **Most appear to arise from pluripotent mesenchymal stem cells** – acquire somatic “driver” mutations
- 20% have simple karyotypes - e.g., Synovial sarcoma, Ewing sarcoma
- 80% have complex karyotypes – multiple chromosomal gains and losses, e.g., undifferentiated pleomorphic sarcoma
- Pathological classification combines **morphology, IHC, and molecular diagnostics; grade and stage – important prognostic indicators**

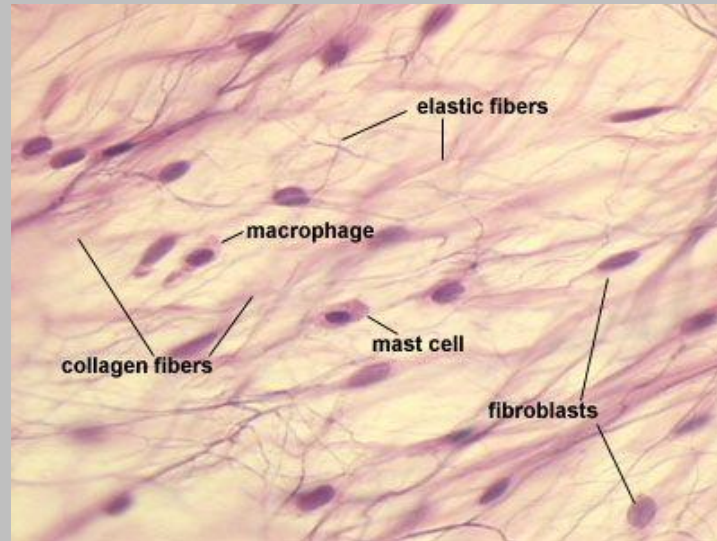
Tumor	Cytogenetic Abnormality	Gene Fusion	Proposed Function
Ewing sarcoma	t(11;22) (q24;q12) t(21;22)(q22;q12)	<i>EWS–FLI1</i> <i>EWS–ERG</i>	Disordered protein with multiple functions, including aberrant transcription, cell cycle regulation, RNA splicing, and telomerase
Extraskeletal myxoid chondrosarcoma	t(9;22)(q22;q12)	<i>EWS–CHN</i>	
Desmoplastic small round-cell tumor	t(11;22)(p13;q12)	<i>EWS–WT1</i>	
Clear-cell sarcoma	t(12;22)(q13;q12)	<i>EWS–ATF1</i>	
Liposarcoma—myxoid type	t(12;16)(q13;p11)	<i>FUS–DDIT3</i>	Arrests adipocytic differentiation
Synovial sarcoma	t(x;18) (p11;q11)	<i>SS18–SSX1</i> <i>SS18–SSX2</i> <i>SS18–SSX4</i>	Chimeric transcription factors, interrupts cell cycle control
Rhabdomyosarcoma—alveolar type	t(2;13) (q35;q14) t(1;13) (p36;q14)	<i>PAX3–FOXO1</i> <i>PAX7–FOXO1</i>	Chimeric transcription factors, disrupts skeletal muscle differentiation
Dermatofibrosarcoma protuberans	t(17;22)(q22;q15)	<i>COLA1–PDGFB</i>	Promoter driven overexpression of PDGF- β , autocrine stimulation
Alveolar soft-part sarcoma	t(X;17)(p11.2;q25)	<i>TFE3–ASPL</i>	Unknown
Infantile fibrosarcoma	t(12;15)(p13;q23)	<i>ETV6–NTRK3</i>	Chimeric tyrosine kinase leads to constitutively active Ras/MAPK pathway
Nodular fasciitis	t(22;17)	<i>MYH9–USP6</i>	Increased Wnt/ β -catenin signaling

Category	Behavior	Tumor Type	Common Locations	Age (years)	Morphology
Adipose	Benign	Lipoma	Superficial extremity, trunk	40–60	Mature adipose tissue
	Malignant	Well-differentiated liposarcoma	Deep extremity, retroperitoneum	50–60	Adipose tissue with scattered atypical spindle cells
		Myxoid liposarcoma	Thigh, leg	30s	Myxoid matrix, “chicken wire” vessels, round cells, lipoblasts
Fibrous	Benign	Nodular fasciitis	Arm, forearm	20–30	Tissue culture growth, extravasated erythrocytes,
		Deep fibromatosis	Abdominal wall	30–40	Dense collagen, long, unidirectional fascicles
Skeletal muscle	Benign	Rhabdomyoma	Head and neck	0–60	Polygonal rhabdomyoblasts, “spider” cells
	Malignant	Alveolar rhabdomyosarcoma	Extremities, sinuses	5–15	Uniform round discohesive cells between septae
		Embryonal rhabdomyosarcoma	Genitourinary tract	1–5	Primitive spindle cells, “strap” cells
Smooth muscle	Benign	Leiomyoma	Extremity	20s	Uniform, plump eosinophilic cells in fascicles
	Malignant	Leiomyosarcoma	Thigh, retroperitoneum	40–60	Pleomorphic eosinophilic cells
Vascular	Benign	Hemangioma	Head and neck	0–10	Circumscribed mass of capillary or venous channels
	Malignant	Angiosarcoma	Skin, deep lower extremity	50–80	Infiltrating capillary channels
Nerve sheath	Benign	Schwannoma	Head and neck	20–50	Encapsulated, fibrillar stroma, nuclear palisading
		Neurofibroma	Wide, cutaneous, subcutis	10–20+	Myxoid, ropy collagen, loose fascicles, mast cells
	Malignant	Malignant peripheral nerve sheath tumor	Extremities, shoulder girdle	20–50	Tight fascicles, atypia, mitotic activity, necrosis
Uncertain histotype	Benign	Solitary fibrous tumor	Pelvis, pleura	20–70	Branching ectatic vessels
	Malignant	Synovial sarcoma	Thigh, leg	15–40	Tight fascicles of uniform basophilic spindle cells, pseudoglandular structures
		Undifferentiated pleomorphic sarcoma	Thigh	40–70	High-grade anaplastic polygonal, round, or spindle cells, bizarre nuclei, atypical mitoses, necrosis
Robbins and Cotran, Pathologic Basis of Disease, 10th ed. 2020, Ch.26, Table 26.8					

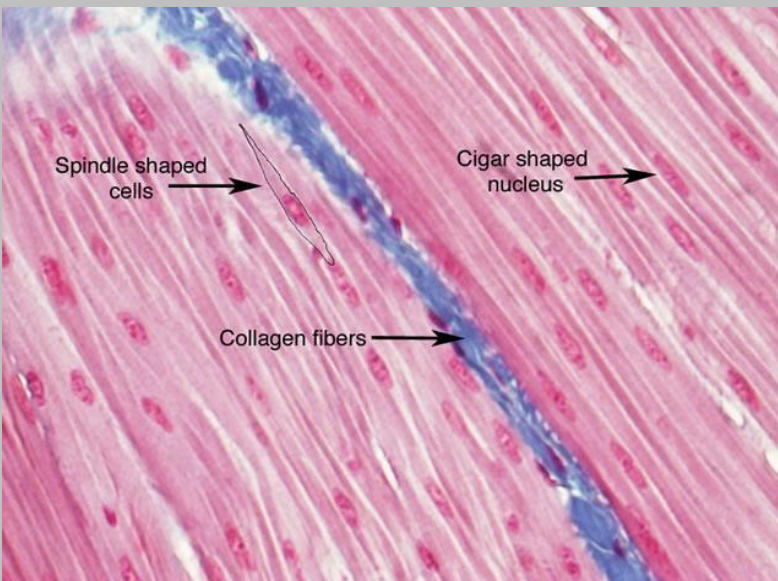
Soft tissue cells – Normal histology



<https://www.pathologyoutlines.com/imgau/softtissueadiposewhitefatule01.jpg>

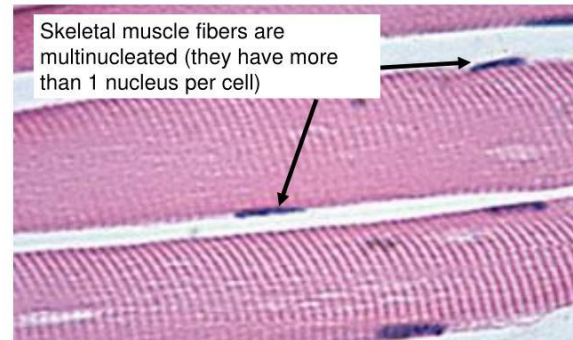


<https://i.pinimg.com/736x/01/0d/0f/010d0fd35aa582a2cfc7989795365c42--loose-connective-tissue-anatomy-and-physiology.jpg>



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Skeletal Muscle Long Section: Multinucleated



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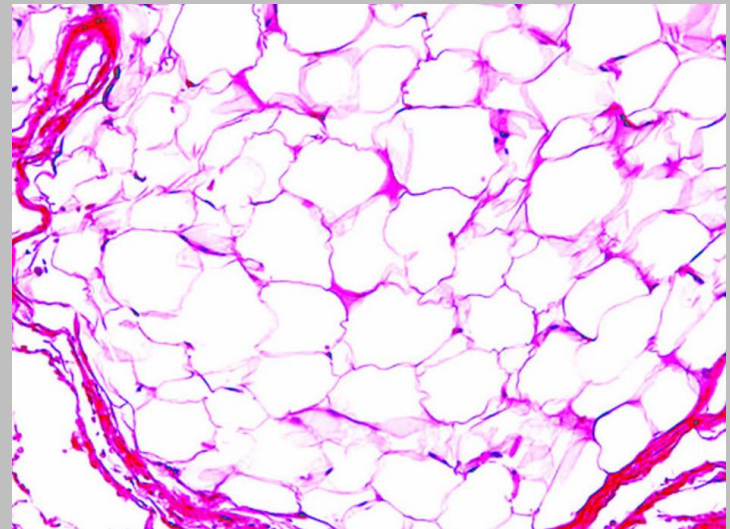
LIPOMA

- **Benign** tumor of fat (**adipose** tissue); **most common** soft tissue tumor in **adults**; rare variants
- **Well-encapsulated** mass of **mature** adipocytes; usually in subcutis of proximal extremities and trunk during **middle adulthood**
- Infrequently: Large, intramuscular, and poorly circumscribed
- Lipomatosis: **Multifocal** lipomas involving a limb
- Most are **soft, mobile, painless**, cured by simple excision

LIPOMA



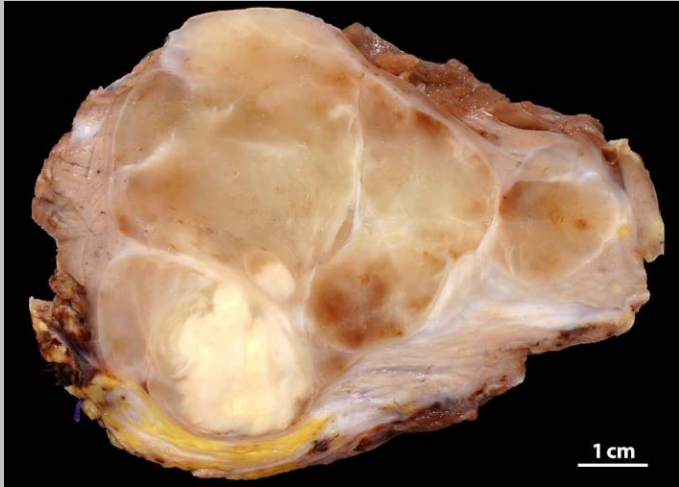
Well circumscribed subcutaneous lipoma with fibrous capsule



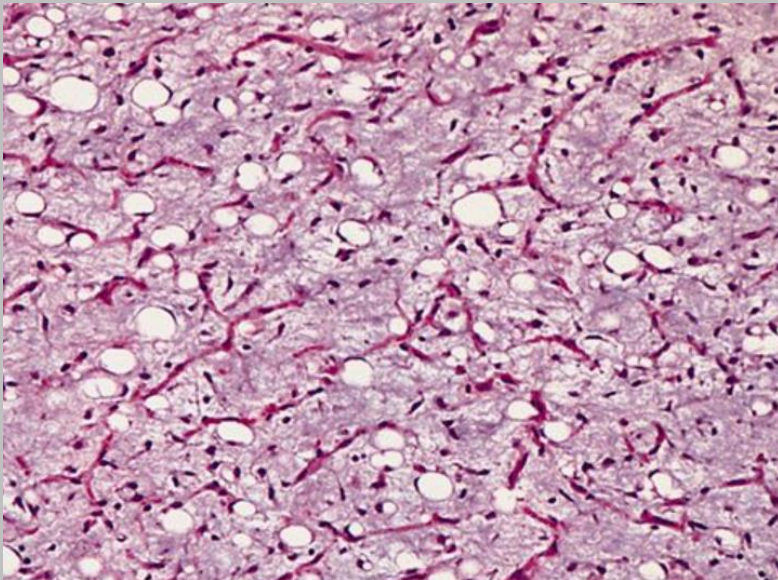
LIPOSARCOMA

- **Malignant tumors of adipose tissue; most common sarcomas of adulthood**
- Usually in **deep soft tissues** of proximal **extremities** and **retroperitoneum** of **6th and 7th decades** of life with **pain, soft, swelling, fatigue, decreased of range of motion**
- Three distinct subtypes: well-differentiated, myxoid, and pleomorphic; have different genetic aberrations
 - **Well-differentiated liposarcomas** have amplifications of chromosome region 12q13–q15, which includes the p53 inhibitor MDM2
 - **Myxoid liposarcoma** have fusion gene generated by a t(12;16) translocation, arrests adipocyte differentiation, leading to unregulated proliferation of primitive cells
 - **Pleomorphic liposarcomas** have complex karyotypes without reproducible genetic abnormalities
- **Well-differentiated liposarcoma**: Mature adipocytes with scattered **atypical spindle cells**
- **Myxoid liposarcoma**: Abundant **basophilic** extracellular **matrix**, arborizing **capillaries**, and primitive cells at various stages of adipocyte differentiation
- **Pleomorphic liposarcoma**: Pleomorphic spindle cells of high grade sarcoma, and anaplastic cells with bizarre nuclei mixed with occasional immature adipocytes (**lipoblasts**), areas of necrosis and hemorrhage, containing less fatty tissue than well-differentiated, usually well demarcated but not encapsulated; stains for **S-100**, negative for **keratins, desmin**.
- Recur locally, unless adequately excised
- Well-differentiated tumors – indolent; **pleomorphic liposarcomas** – **aggressive, frequently metastasize**; myxoid tumors behavior intermediate

Myxoid



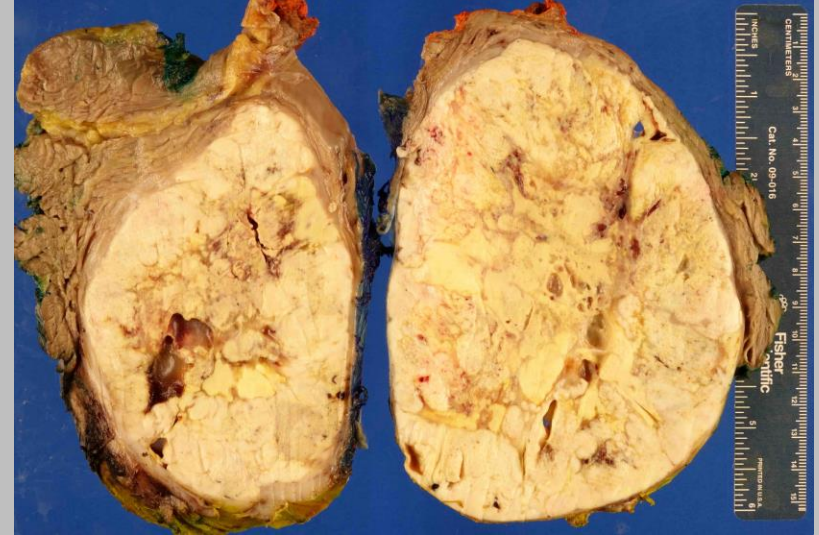
<https://v2.pathorama.ch/storage/samples/008589.jpg>
Lobulated, circumscribed gray white tumor within skeletal muscle



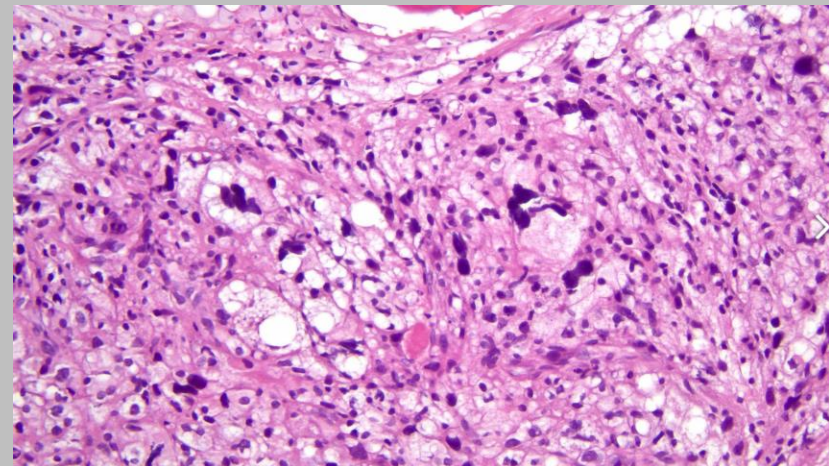
Low grade myxoid liposarcoma showing delicate capillary network
<https://www.ncbi.nlm.nih.gov/pubmed/>

Pleomorphic

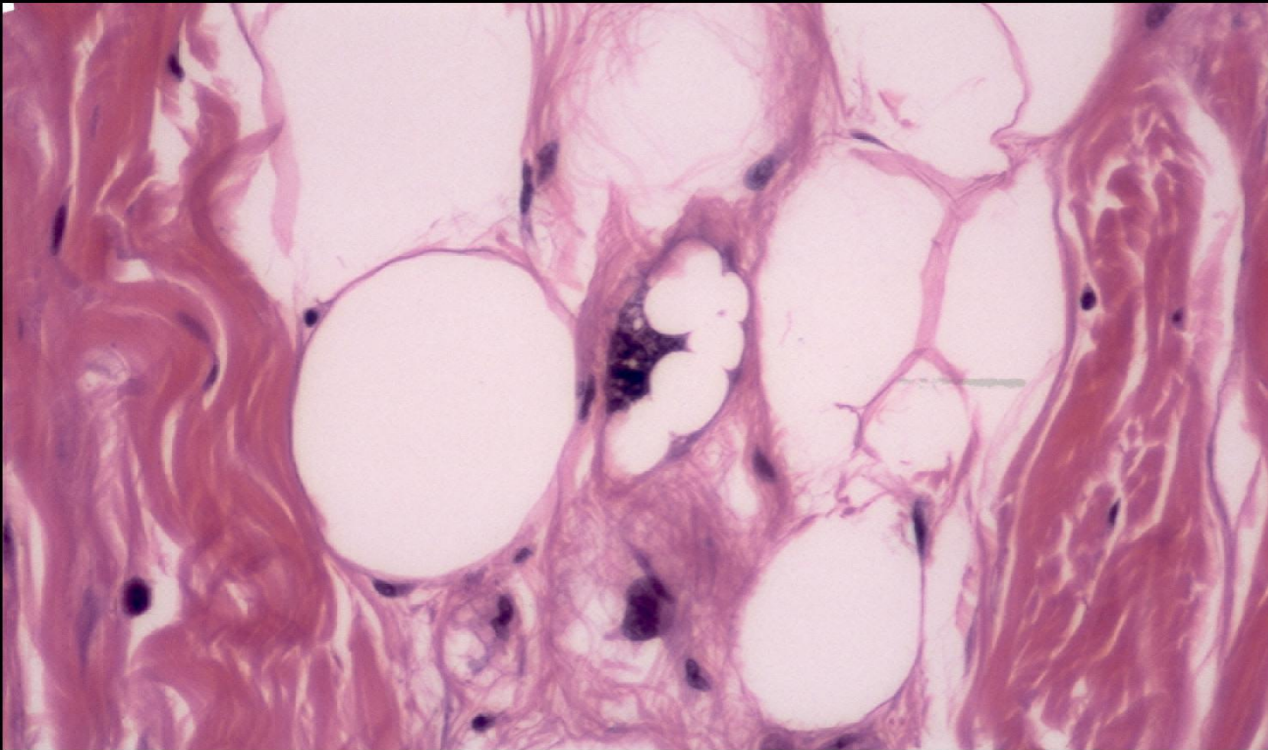
<https://www.ncbi.nlm.nih.gov/pubmed/>



Within skeletal muscle. Focal central areas with cystic degeneration and necrosis (yellow discolored areas), focal areas of adjacent tissue invasion



Pleomorphic cell population with scattered lipoblastic features



Lipoblast, with
scalloped-
shaped nucleus



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[European Journal of Dermatology. Volume 15,
Numéro 1, 43-5, January-February 2005,
Clinical report](#)

Question

- A large fatty tumor is removed from the retroperitoneum of a 55 year old woman. The tumor is malignant and derived from adipose tissue and is diagnosed as a pleomorphic liposarcoma. Which of the following best describes this malignancy?
- A. Histologically it will show a myxoid background
- B. Lipoblasts are likely to be present histologically
- C. This tumor is likely to be cured by simple excision
- D. This is a tumor of well-encapsulate mature adipocytes
- E. No genetic aberrations are likely to be present

Answer

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NODULAR FASCIITIS

- **Self-limited fibroblastic and myofibroblastic proliferation**, occurs in **young** adults; most often forearm, chest, and back; **feels firm, fixed, rubbery, and solid, may be tender or painful to touch, rapid growth over weeks to months**
- History of trauma in 10% to 50% of cases, (once thought to be reactive)
- Clonal proliferation with t(17;22) translocation - produces a MYH9-USP6 fusion
- Unknown why neoplastic cells do not become malignant
- **Usually regresses spontaneously; if excised, rarely recurs**
- **Deep dermis, subcutis, fascia, or muscle**

Robbins and Cotran, Pathologic Basis of Disease 10th ed. 2020, Ch.26

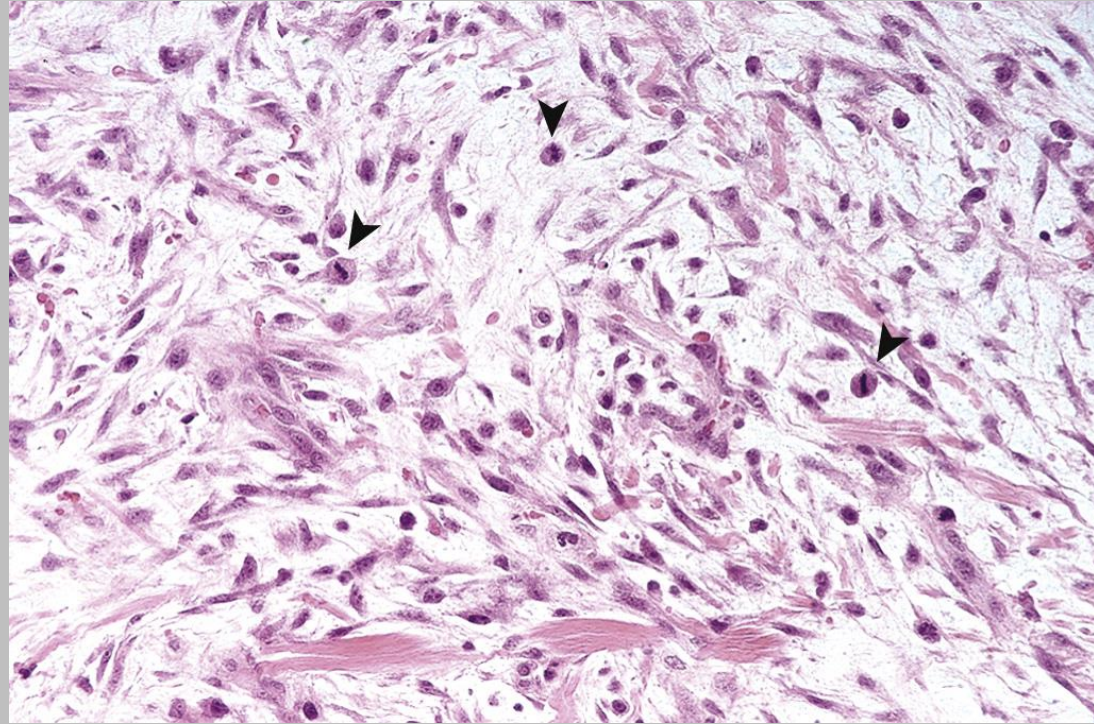
Nodular Fascitis

- Gross: non-encapsulated lesion, well-circumscribed or slightly infiltrative, < 3 cm
- Histo: Plump, immature-appearing **fibroblasts** and myofibroblasts with elongated nuclei showing punctate nucleoli; **frequent mitoses – none abnormal**
- Changes from hypercellular regions with **myxoid** stroma to hypocellular areas with **fibrous** stroma (zonation)
- Storiform or fascicular patterns, common in cellular areas; metaplastic bone, cystic areas, prominent vessels, extravasated red cells, and infiltrating lymphocytes common

Nodular fasciitis

- Plump, randomly oriented spindle cells surrounded by myxoid stroma; “tissue culture” appearance

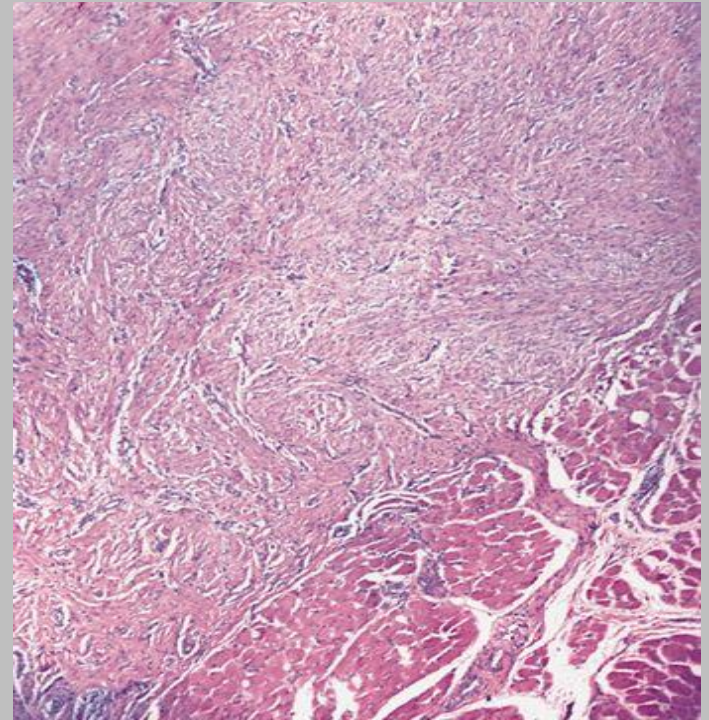
Note mitotic activity (arrowheads) and extravasated red blood cells



Robbins and Cotran, Pathologic Basis of Disease 10th ed. 2020, Ch.26, Fig. 26.51

Superficial Fibromatoses

- **Benign growth** - can cause local **deformity, painless, firm, subcutaneous induration**, innocuous clinical course
- M > F males
- Nodular, proliferative lesions made of **plump spindle cells (fibroblasts) arranged in poorly defined long fascicles surrounded by abundant dense collagen**
- Clinical subtypes:
 - **Dupuytren contracture (palmar fibromatosis)**: Irregular or nodular thickening of superficial palmar fascia, unilateral or bilateral; increased incidence with age; attachment to overlying skin: puckering or dimpling, slowly progressive flexion contracture
 - **Ledderhose disease (plantar fibromatosis)**: In boys < 10 years of age into adolescence; unilateral, does not cause contractures (may be associated with palmar and penile fibromatosis)
 - **Peyronie disease: (penile fibromatosis)**: Palpable induration or mass on dorsolateral aspect of penis; may cause abnormal curvature of the shaft and constriction of the urethra
- Palmar and plantar progress in about 50% of cases; some resolve spontaneously; recurrence is common (even after excision)

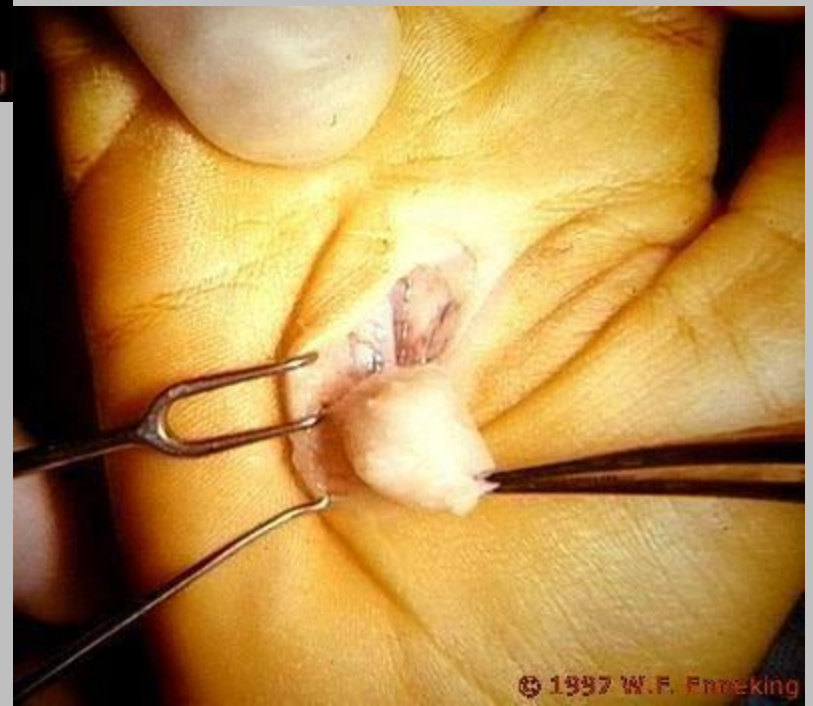


Superficial Palmar fibromatosis

<http://www.pathologyoutlines.com/topic/softtissuefibromatosis/superficial.html>

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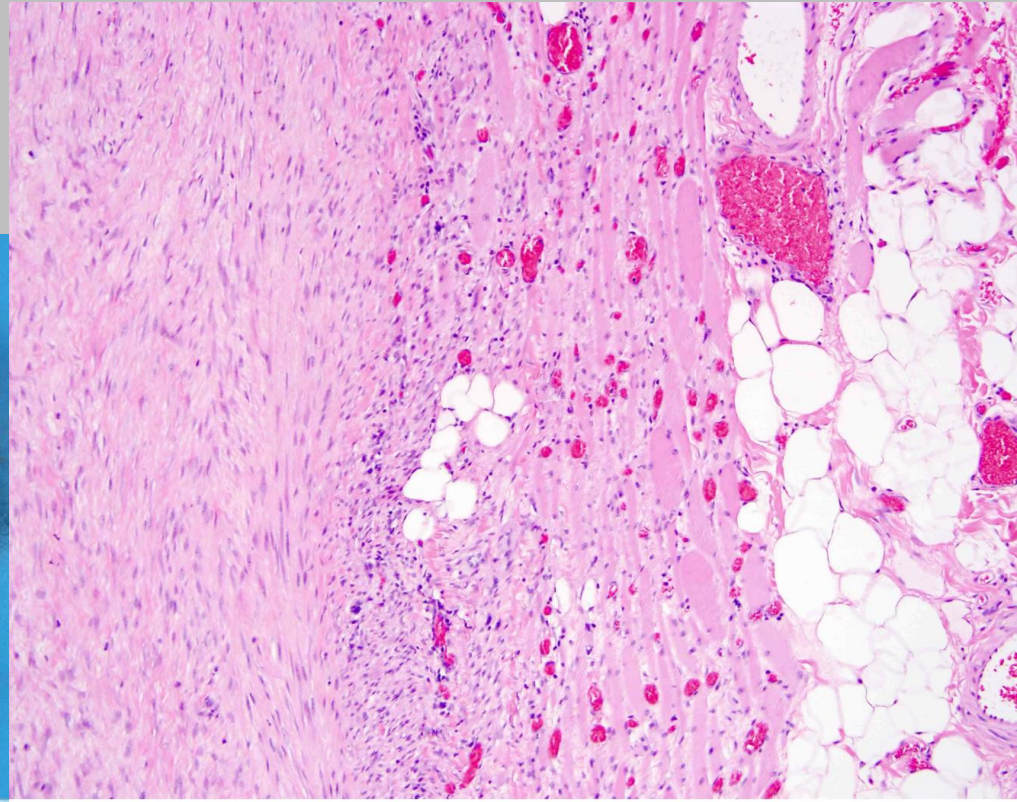
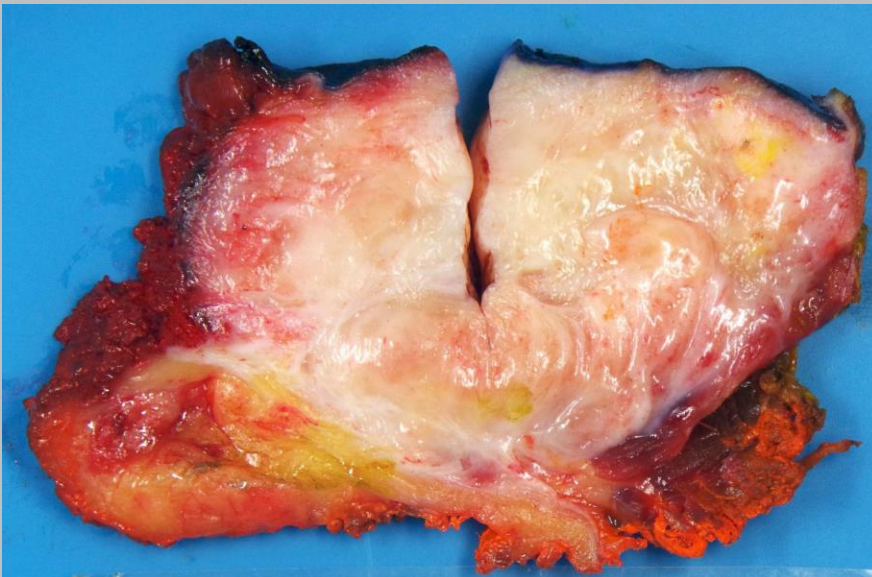
Deep Fibromatoses (Desmoid Tumors)

- Large (avg. ~ 10 cm), infiltrative masses, **frequently recur but do not metastasize**; teens to 30s, mostly **women**
- **Abdominal** fibromatosis usually arises in musculoaponeurotic structures of anterior lateral abdominal wall; may also be in limb girdles or mesentery
- **Deep fibromatoses contain mutations in the APC or CTNNB1 (β -catenin) genes, both lead to increased Wnt signaling**
- Majority have sporadic CTNNB1 mutations (note: individuals with familial adenomatous polyposis (germline APC mutations, Gardner syndrome) are predisposed to deep fibromatosis)

Deep Fibromatosis

- Gray-white, firm, poorly demarcated masses from 1 to 15 cm
- **Rubbery**, tough, fixed, **infiltrate** surrounding muscle, nerve, and fat
- May be disfiguring or disabling, **painful**
- Micro: Bland fibroblasts arranged in long parallel fascicles amid dense collagen; no cellular; atypia; no mitoses; resembles scar tissue
- Lesions are infiltrative, complete excision difficult, local **infiltration** into adjacent muscle; **do not spread** to other areas of the body (but they can be very aggressive, acting more like cancers and growing into nearby structures and organs, so are often cared for by cancer doctors)

Deep fibromatosis- Cut surface of lesion shows lack of encapsulation, diffuse infiltration into the adjacent fat, deep fascia, and muscle



tumors may show extension of thin strands of fibrous tissue into surrounding fat and muscle

infiltration of surrounding muscle and soft tissue may be present at the border

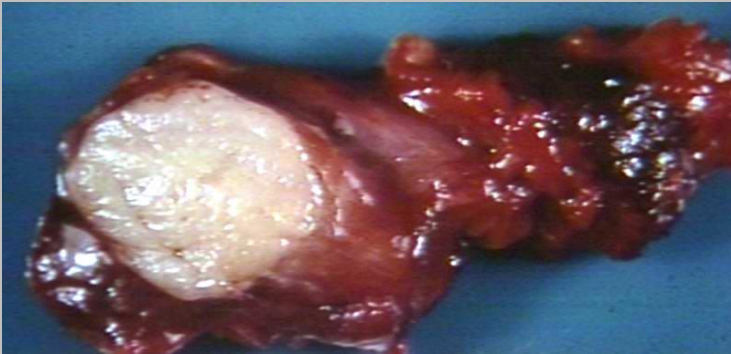
<https://www.pathologyoutlines.com/imgau/softtissuefibromatosisdeepobeidingross02update.jpg>

<https://www.ncbi.nlm.nih.gov/pubmed/>

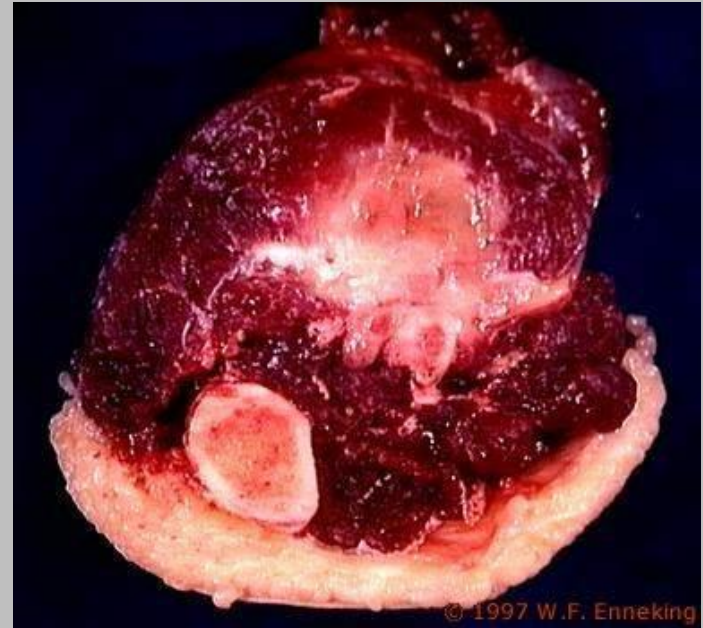
Fibrosarcoma

- **Rare, malignant tumor of fibroblasts** ($\leq 3\%$ of adult sarcomas); most patients 40 - 55 years
- Two types of fibrosarcomas: infantile/congenital-type fibrosarcoma (rarely mets), and adult-type fibrosarcoma
- Many cases formerly called fibrosarcoma where actually dedifferentiated liposarcoma, fibromatosis, fibrosarcomatous DFSP, low-grade fibromyxoid sarcoma, MPNST, synovial sarcoma or MFH-pleomorphic
- Usually **deep soft tissue** of **lower extremities or trunk**, **painful, fixed, soft** tumor, **unexplained weight loss**
- 50% recur, 25% metastasize (lung, bone); more metastases if more cellular and higher mitotic activity
- Survival: 5 year - 41%, 10 year - 29%
- **Diagnosis of exclusion:** IHC and molecular studies identify subtypes
- May appear well circumscribed but non-encapsulated; fleshy, hemorrhagic, necrotic, white-tan
- **Histo: Highly cellular fibroblastic proliferation in herringbone pattern and variable collagen;** scant cytoplasm, tapering elongated dark nuclei with increased granular chromatin, variable nucleoli; **mitotic activity present, often with abnormal forms, stains positive for vimentin, negative for S-100, cytokeratins**
- Treatment : Radical excision, radiation if residual tumor or positive margins; possibly chemotherapy if high grade
- Definitive cause not identified: Variable genetic mutations; predisposing factors-sites of prior injury, chronic inflammation, irradiated tissue, preexisting well-differentiated liposarcoma

Fibrosarcoma



Gross: “fish-flesh” surface



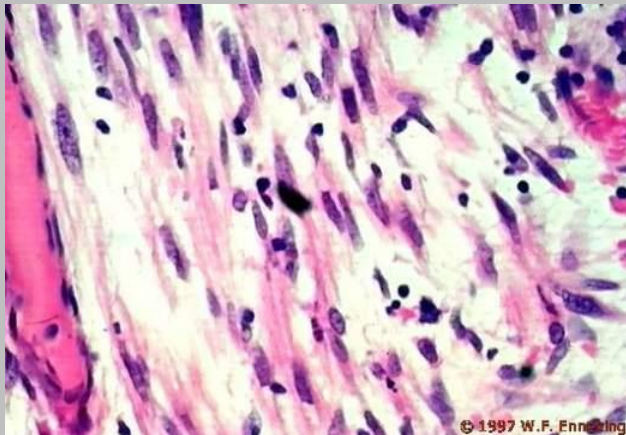
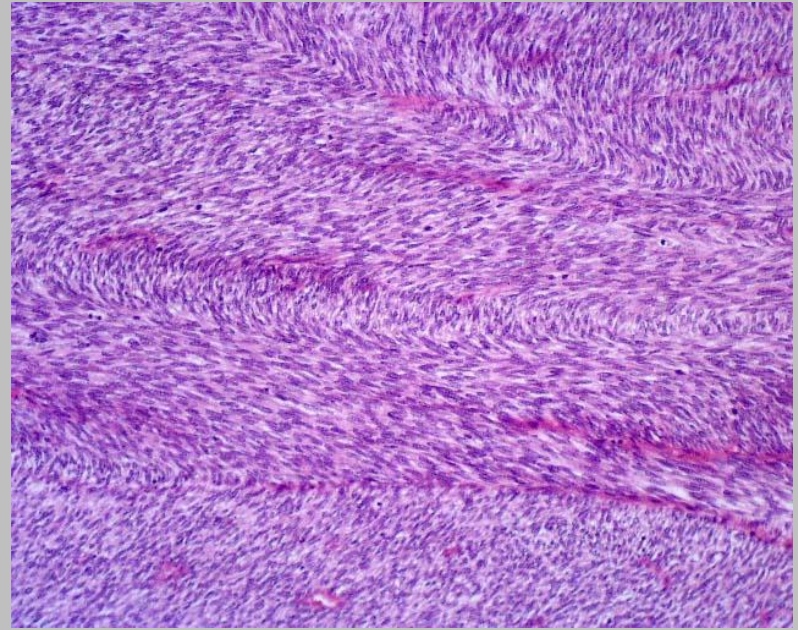
Gross: Infiltrative lesions, “pseudo-encapsulated”, cut section shows white rope-like appearance with lobulation, hemorrhage and necrosis

Fibrosarcoma microscopic

Cellular, spindled hyperchromatic cells with increase nuclear to cytoplasmic ratio; **herring bone pattern (classical)**



“herring bone” pattern

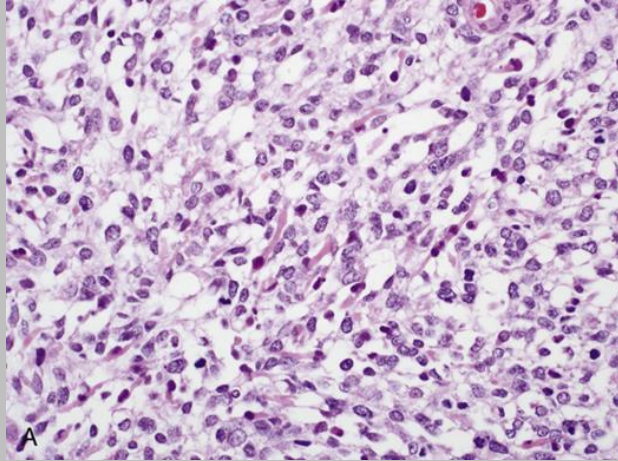


Hyperchromatic nuclei with increased N/C ratio

Rhabdomyosarcoma (RMS)

- **Rare malignant mesenchymal tumor with skeletal muscle differentiation** (But most common soft tissue sarcoma of childhood, cancer in the connective tissue wherein the mesenchymal-derived malignant cells resemble the primitive developing skeletal muscle, usually starts in muscle tissue)
- Four subtypes are recognized: **alveolar (20%), embryonal (50%), pleomorphic (20%)**
- Generally considered to be a disease of **childhood**, as the vast majority of cases occur in those below the age of 18
- Symptoms vary depending on the tumor's location, commonly a **quickly growing mass with pain, unexplained weight loss**
- **Alveolar and embryonal subtypes: Most common soft tissue sarcomas of childhood and adolescence**, usually before 20 years of age (**alveolar**, older children and young adults, **20-35 years-old, arms, legs, or trunk**)
 - **Pediatric rhabdomyosarcomas** often occur in the **sinuses, head and neck, and genitourinary tract**
- **Pleomorphic rhabdomyosarcoma usually occurs in adults** (typically affects adults ages 50 and older); spindle cell/sclerosing type affects all ages, most common locations lower and upper **limbs**
- **Aggressive** and usually treated by **surgical** resection and **chemotherapy**, with or without **radiation** therapy
- Histologic type and location correlated with survival; **sarcoma botryoides (subtype of embryonal rhabdomyosarcoma) has the most favorable prognosis with good results from treatment**; **alveolar** rhabdomyosarcoma four year survival rate without remission for local tumors is 65 percent, while the four year survival rate with **metastatic is only 15%**; **pleomorphic** rhabdomyosarcoma often **fatal**
- Stains positive for **desmin, myogenin**, negative for **S-100, cytokeratins**. Presence of the **PAX-FOXO1 fusion gene** that is found in some patients with RMS is associated with **poorer survival** (due to t(2;13) or t(1;13) translocations, PAX3 is a transcription factor that initiates skeletal muscle differentiation, PAX3-FOXO1 fusion protein interferes with differentiation)

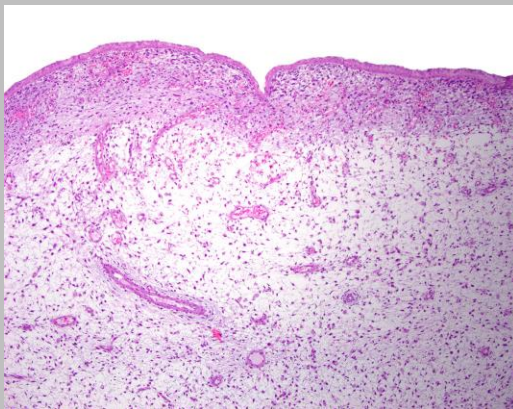
Embryonal rhabdomyosarcoma



Robbins and Cotran, Pathologic Basis of Disease, 10th ed.
2020, Ch.26, Fig. 26.53



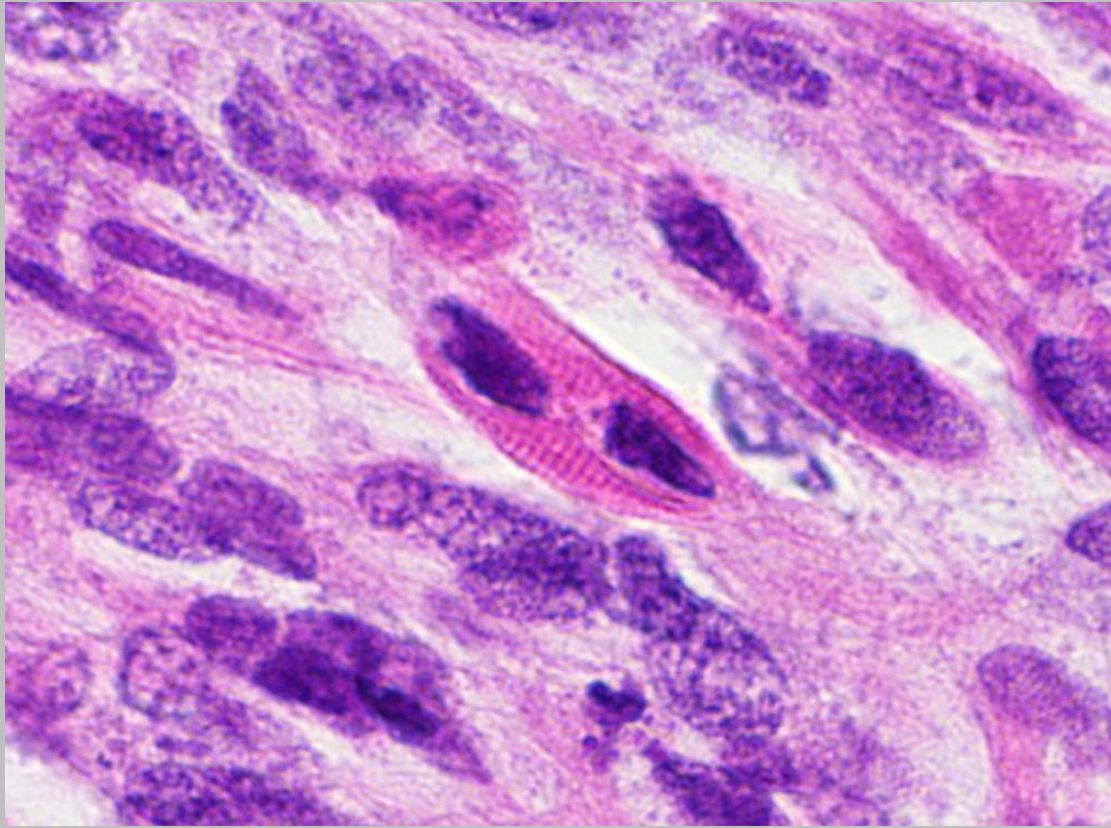
<https://i.ytimg.com/vi/dz-T94625Bw/maxresdefault.jpg>



- Soft, gray, infiltrative mass
- Embryonal subtype: Cells ranging from **primitive** and round to densely eosinophilic with **skeletal muscle differentiation**; tumor cells recapitulate skeletal muscle at various stages of differentiation, include sheets of primitive round and spindled cells in myxoid stroma
- **Rhabdomyoblasts** with visible cross-striations may be present
- **Sarcoma botryoides: embryonal rhabdomyosarcoma variant** that develops in the walls of hollow, mucosal-lined structures (e.g., nasopharynx, common bile duct, bladder, and vagina)
- Grows in polypoid fashion, “**clusters of grapes**” in hollow organ (e.g, bladder or vagina)
- Cells of the botryoid variant are defined by a dense tumor layer under an epithelium (**cambium layer**)
- Sarcoma botryoides subtype has **the best prognosis**

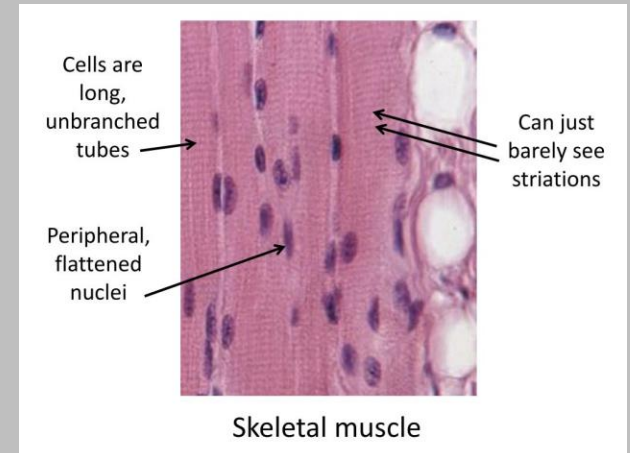
Cambium layer,
<https://www.pathologyoutlines.com/imgau/cervixrhabdomyosarcomaDevins2.jpg>

Rhabdomyoblast



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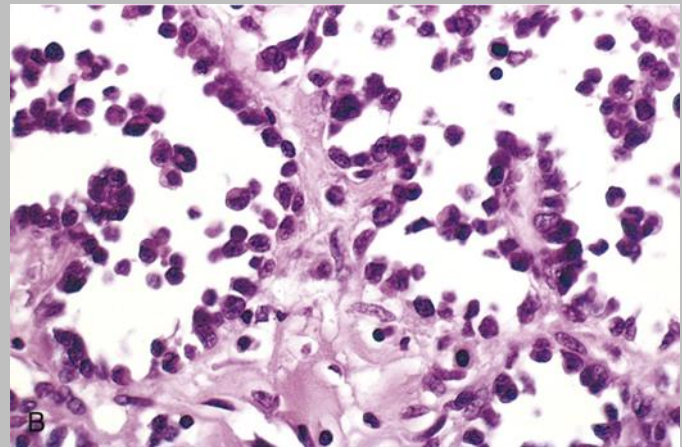
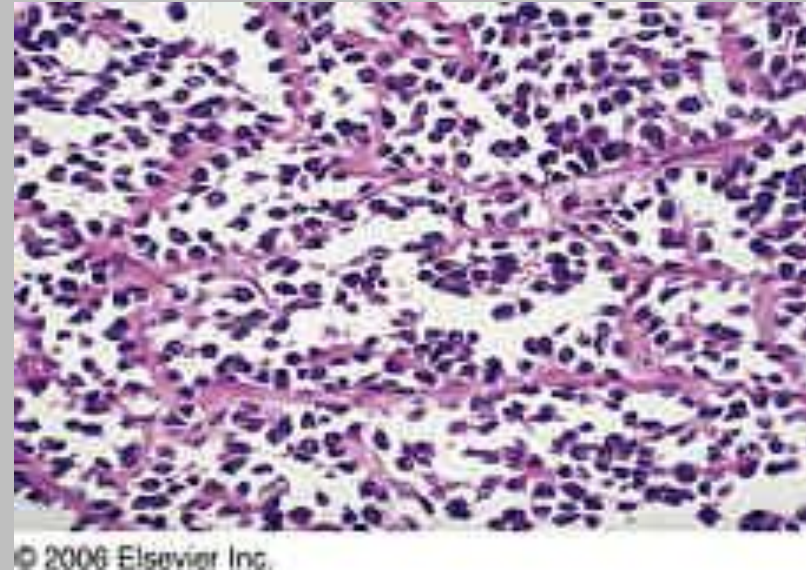
Normal



<https://image2.slideserve.com/3835151/skeletal-muscle2-l.jpg>

Alveolar Rhabdomyosarcoma

- Micro: Crossed by network of fibrous septae that divide cells into clusters or aggregates, creating crude resemblance to pulmonary alveoli
- Tumor cells: uniform and round with little cytoplasm; cross-striations uncommon

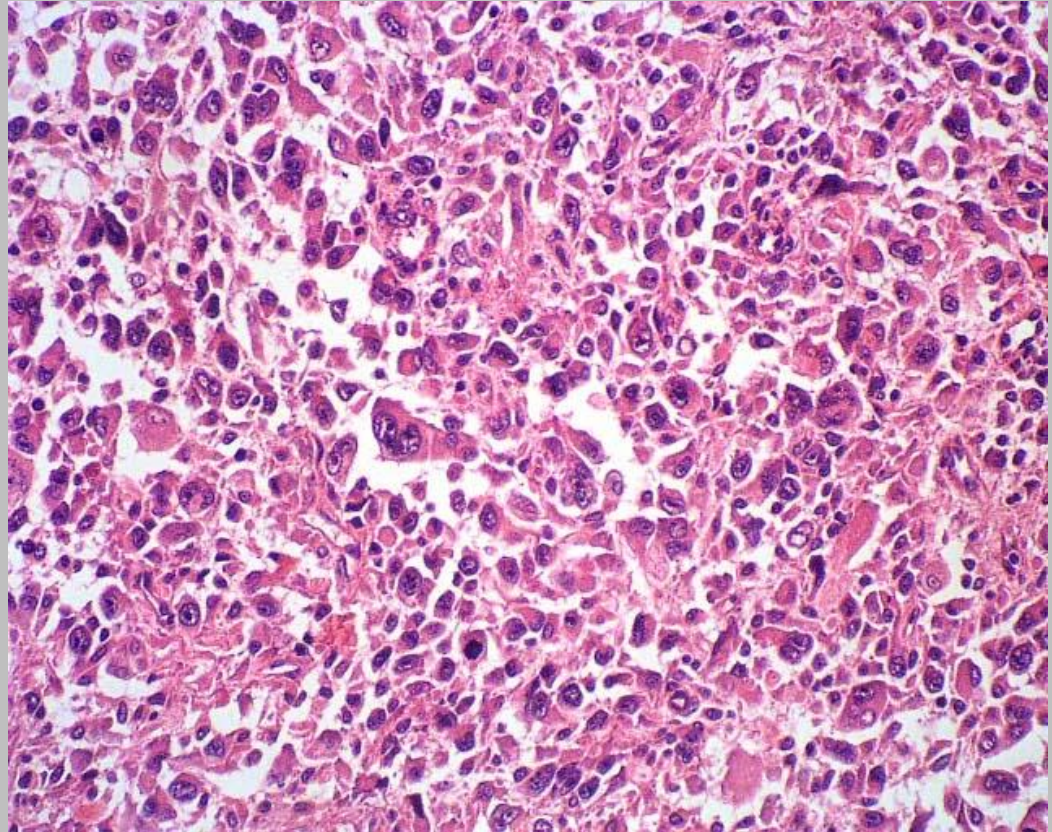


Numerous spaces lined by discohesive, uniform round tumor cells

Robbins and Cotran, Pathologic Basis of Disease, 10 th ed., 2020, Ch.26, Fig. 26.53

Pleomorphic Rhabdomyosarcoma

- Micro: Large, sometimes multinucleated, bizarre eosinophilic tumor cells; can overlap histologically with other pleomorphic sarcomas
- Immunohistochemistry (e.g., myogenin, desmin staining) usually needed to confirm rhabdomyoblastic differentiation



Question

A 6 year old girl has a polypoid mass growing from her vagina which resembles a grape-like cluster of tissue. This mass is a malignant skeletal muscle tumor. What is most likely a feature of this neoplasm?

- A. Microscopic examination reveals areas which crudely resemble lung alveoli
- B. There are not translocations present
- C. Prognosis is better than other malignant skeletal muscle tumors
- D. Immunohistochemistry is needed to confirm fibroblast differentiation
- E. This is the least common soft tissue sarcoma in children

Answer

A 6 year old girl has a polypoid mass growing from her vagina which resembles a grape-like cluster of tissue. This mass is a malignant skeletal muscle tumor. What is most likely a feature of this neoplasm?

- A. Microscopic examination reveals areas which crudely resemble lung alveoli
- B. There are no translocations present
- C. **Prognosis is better than other malignant skeletal muscle tumors**
- D. Immunohistochemistry is needed to confirm fibroblast differentiation
- E. This is the least common soft tissue sarcoma in children

Synovial sarcoma

- The most common location for these tumors is within the **deep soft tissues adjacent to large joints**, e.g. knee and popliteal fossa
- Can present in locations that **lack synovium; uncertain origin**, inconsistent with origin from synoviocytes or synovial structures, e.g. joints, tendon sheaths and bursae
- Approximately 10% of all soft tissue sarcomas, usually **20s to 40s**
- Individuals often present with **deep-seated pain, numbness, limited range of motion**, firm, fixed, lump or **swelling** under the skin, often **around the joints** in the arms or legs
- Symptoms similar to those of arthritis or bursitis, such as swelling and inflammation present for **several years**
- Characteristic chromosomal translocation **t(x;18)(p11;q11)**: Produces fusion of SS18 gene to one of three SSX genes - fusions encode chimeric proteins that interfere with normal chromatin remodeling (the repositioning of nucleosomes on chromatin that influences gene expression)

Synovial Sarcoma

- Micro: Monophasic or biphasic
- Monophasic - uniform **spindle cells** with scant cytoplasm and dense chromatin arranged in short, tightly packed fascicles and **biphasic also contains gland-like structures composed of cuboidal to columnar epithelioid cells**
- **IHC helpful because tumor cells, especially in biphasic type, are positive for epithelial antigens versus most other sarcomas (e.g., cytokeratins, epithelial component is consistently positive for CK7, CK8, CK14, CK18, CK19; spindle cell component usually focal and variably positive for CK7, CK19, CK18, CK8, CK14, CK17, CK20)**
- Treated aggressively with limb-sparing surgery and chemotherapy
- 5-year survival varies from 25% to 62%, related to tumor stage and patient age
- In general, synovial sarcoma has a survival rate of 50-60% at 5 years and 40-50% at 10 years
- The survival rate for synovial sarcoma is lower in cases where the cancer has returned or spread
- Treatment involves surgery, chemotherapy and radiation. Some factors for prognosis include the following: Small tumor size—less than 5cm wide, few mitoses, easy to remove by surgery
- Common sites of metastases are the lung and, **unusually for sarcomas, regional lymph nodes**

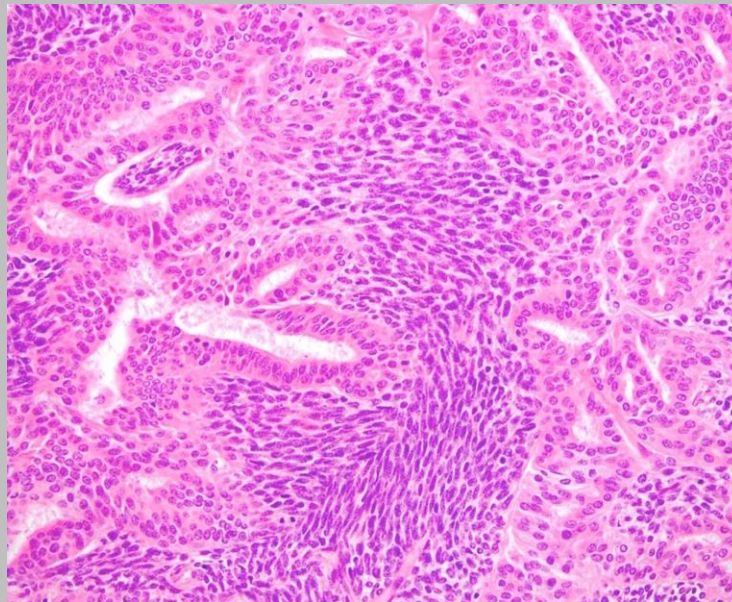
Synovial Sarcoma



MRI Coronal Synovial Sarcoma of Distal Thigh Knee Area

<https://tumorsurgery.org/portals/0/Synovial%20Sarcoma/Figure%203A%20MRI%20Coronal%20Synovial%20Sarcoma%20of%20Distal%20Thigh%20Knee%20Area.jpg>

- Synovial sarcoma with classic **biphasic** spindle cell and gland-like histologic appearance; glands are composed of **closely packed round to ovoid cells**

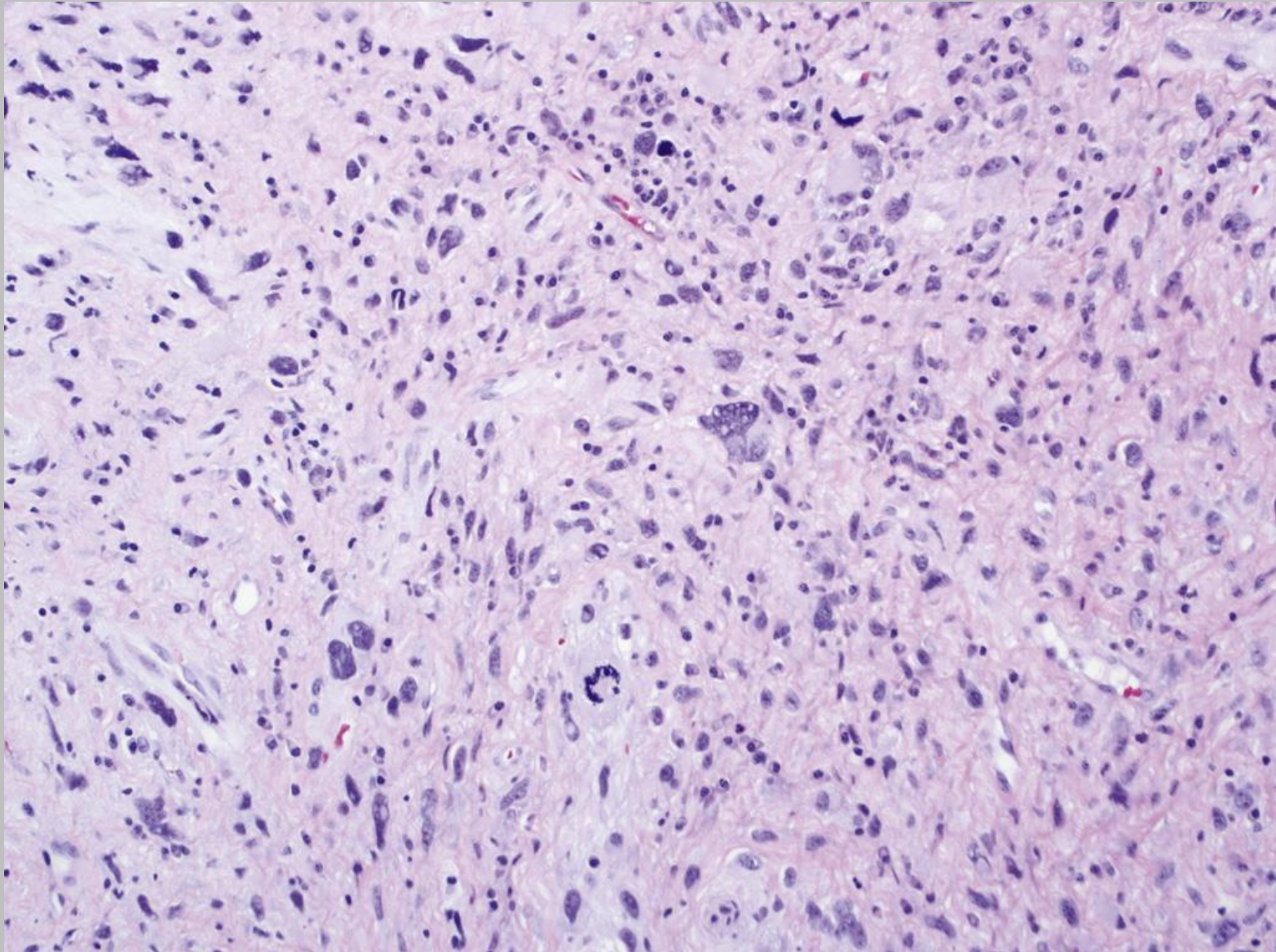


<https://www.ncbi.nlm.nih.gov/pubmed/>

Undifferentiated pleomorphic sarcoma

- Include **malignant mesenchymal tumors with high-grade, pleomorphic cells that cannot be classified into another category by histomorphology, immunophenotype, and genetics** (old name malignant fibrous histiocyoma)
- Most arise in the **deep soft tissues of extremities**, especially **thigh** of **middle-aged** or older adults as a growing fixed, firm, lump or area of swelling. If it grows very large, there may be **pain**, tingling and numbness.
- Usually large, gray-white fleshy masses; can grow to 20 cm
- Hemorrhage common
- Micro: **Extremely pleomorphic, composed of sheets of large, anaplastic spindled to polygonal cells with hyperchromatic irregular, sometimes bizarre nuclei; mitotic figures, including atypical asymmetric forms abundant**
- **By definition, tumor cells lack both differentiation along recognized lineages phenotypically (stains) and characteristic genetic defects**
- Aggressive malignancies: Treated with surgery and adjuvant chemotherapy and/or radiation; prognosis is generally poor. Metastases occur in 30% to 50% of cases

Undifferentiated pleomorphic sarcoma showing anaplastic spindled to polygonal cells



Other soft tissue tumors

- Rhabdomyoma: Benign skeletal muscle tumor, usually found in heart, associated with tuberous sclerosis
- Leiomyoma: Common benign smooth muscle tumor, commonly in the uterus
- Leiomyosarcoma: Malignant smooth muscle tumor, especially in uterus

Summary Slide

- Lipoma – benign, adipose tissue
- Liposarcoma – malignant, adipose tissue, subtypes well-differentiated, myxoid, pleomorphic
- Nodular fasciitis – benign, fibroblasts
- Superficial dermatoses – palmar, plantar, penile
- Deep fibromatosis – recur, associated with APC and beta-catenin gene versus superficial fibromatosis (both fibroblasts/myofibroblasts)
- Fibrosarcoma – malignant, aggressive, fibroblasts, recur, metastasize, herring bone histological pattern
- Rhabdomyosarcoma – malignant, skeletal muscle
 - Embryonal – subtype Sarcoma Botyoides (best prognosis)
 - Alveolar – t(2;13)
 - Pleomorphic
- Synovial sarcoma – malignant, t(X:18), monophasic or biphasic
- Undifferentiated pleomorphic sarcoma – highly malignant, need special stains and studies,

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